MIL-S-19500/483
AMENDMENT 4
24 October 1985
SUPERSEDING
AMENDMENT 3
11 September 1984

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SILICON, HIGH POWER, THREE PHASE,
FULL WAVE BRIDGE RECTIFIER

This amendment forms a part of MIL-S-19500/483 dated 23 April 1973 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute as printed above.

1.1, delete and substitute the following:

"1.1 Scope. This specification covers the detail requirements for TX, three phase, full wave bridge rectifiers, intended for use in applications at frequencies of 1 KHz or less. The prefix "TX" is used on bridges whose discrete diodes have been submitted to and have passed the special process-conditioning, testing and screening as specified in table IV".

1.3 Maximum ratings. Add footnote reference 3/ to column 1 and the following footnote:

"3/ For JANTX types, add the prefix JX (JXM19500/483-01)."

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* FIGURE 1: Delete and substitute as printed on page 3 of this amendment.

PAGE 4

4.4.2.1.1, delete and substitute:

"4.4.2.1.1 Preburn-in tests (method 1038, test condition A). The parameters I_R and V_F of table IV shall be measured and the data recorded. Other methods may be used to control delta requirements provided they have prior approval from the qualifying activity. All devices that fail to meet the requirements of I_R and V_F in table IV initially shall be removed from the lot and the quantity removed shall be noted on the lot history.

Add "4.4.2.1.2:

"4.4.2.1.2 Classification procedures. The manufacturer may determine the number of cells, initial cell limits, and post burn-in limits for each cell to allow determination of delta requirements. The manufacturer shall maintain adequate records on cell limit values, initial cell distribution data, and the number of rejects per cell. These records shall be retained by the manufacturer for a minimum of two years."

Add 4.4.2.1.3:

"4.4.2.1.3 Classification approval. The manufacturer shall have his classification procedure approved by the qualifying activity before any lot shipments can be made."

Add 4.4.2.1.4:

"4.4.2.1.4 Post burn-in tests. The parameters I_R and V_F of table IV shall be retested within 8 hours after the burn-in test for all devices in the lot. The parameter measured shall not have changed during the burn-in test from the initial value by more than the specified amount in table IV."

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TABLE III, Subgroup 5, LTPD column: Add "10".

Add 4.4.2.4.1.1:

" 4.4.2.4.1.1 Additional mounting conditions for leadless diodes. Leadless diodes may be tested at an elevated ambient burn-in with light mounting clips used as an alternate mounting condition provided the junction temperature is maintained at 160°C to 180°C."

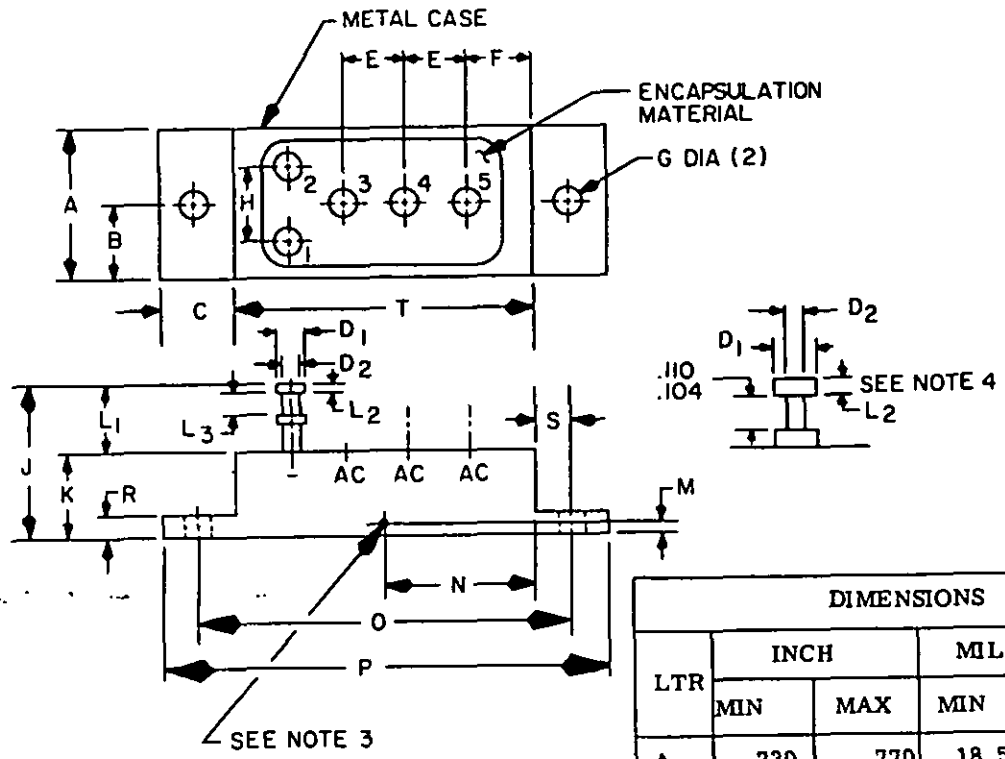
PAGE 8

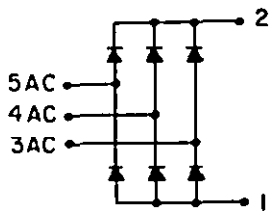
Delete table IV and substitute the following:

TABLE IV. Process and power conditioning discrete diodes for TX level.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
High-temperature (nonoperating)	---	T_A = 200°C; time = 48 hours minimum	---	---	---	---
Thermal shock (temperature cycling)	1051	Test condition F1; time at temperature extremes = 1 minute minimum	---	---	---	---
Hermetic seal	1071	Test condition G or H for fine leaks; test condition C for gross leaks	---	---	---	---
Preburn-in						
Forward voltage	4011	i_F = 39 A(pk); t_p = 8.3 ms; duty cycle ≤ 2%	V_{FM}	---	1.3	V(pk)
Reverse current	4016	DC method; V_R = rated V_{RWM}	I_R	---	1.0	µAdc
Scope display evaluation (see 4.4.3.1)	---		---	---	---	---
Burn-in	---	V_{RWM} = rated; T_A = 25°C; I_o = 3 Adc; t = 164 hrs; f = 60 Hz (see 4.4.2.3, 4.4.2.4 and 4.4.2.4.1)	---	---	---	---
Post burn-in						
End points (within 24 hours of burn-in)						
Forward voltage	4011	i_F = 39 A(pk); t_p = 8.3 ms; duty cycle ≤ 2%	ΔV_F	---	±.05	Vdc
Reverse current	4016	DC method; V_R = rated V_{RWM}	ΔI_R	---	±250	nAdc

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LTR	DIMENSIONS				NOTES
	INCH		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.730	.770	18.54	19.56	
B	.355	.395	9.02	10.03	
C	.355	.395	9.02	10.03	
D_1	.141	.151	3.58	3.84	
D_2	.108	.118	2.74	3.00	
E	.355	.395	9.02	10.03	
F	.230	.270	5.84	6.86	
G	.149	.189	3.78	4.80	
H	.355	.395	9.02	10.03	
J		.82		20.83	
K	.39	.59	9.91	12.95	
L_1		.320	6.10	8.13	
L_2	.015	.030	.38	.76	
L_3	.100	.125	2.54	3.18	
M	.040	.060	1.02	1.52	
N	.72	.78	18.29	19.81	
O	1.84	1.90	46.74	48.26	
P	2.22	2.28	56.39	57.91	
R	.09	.15	2.29	3.81	
S	.168	.208	4.27	5.28	
T	1.47	1.53	37.34	38.86	

NOTES:
1. Terminals shall be tinned.
2. Polarity shall be marked as shown on drawing.
3. Point at which T_C is read (shall be in metal part of case).
4. Optional turret configuration.

FIGURE 1. Semiconductor device, silicon, high power, three phase, full wave bridge rectifier.

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